

Supplementary Material for “Manifold Transfer Learning for Multitask Optimization”

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This supplementary material contains the theoretical interpretation of positive transfer in MTLKT, detailed descriptions of the benchmark suites along with convergence curves across all tasks, navigation trajectories from all 30 independent runs of MTLMTO for the two tasks in test case C4, and a parameter sensitivity analysis including detailed results of parameters n and BCR , as well as a theoretical analysis on the selection of the number of intermediate subspaces n .

I. MTLMTO

A. THEORETICAL INTERPRETATION OF POSITIVE TRANSFER IN MTLKT

MTLKT leverages manifold alignment to capture the geometric connections between tasks, enabling effective and positive KT in EMTO. Nevertheless, it should be noted that positive KT cannot be theoretically guaranteed in all cases, which is consistent with the “no free lunch” theory [S1]. In this section, based on the domain adaptation theory from Ben-David *et al.* [S2], a theoretical interpretation is provided to show that MTLKT is designed to promote positive KT as much as possible. According to [S2], the target error is bounded by three key factors, namely the source error, the source-target distribution discrepancy, and the minimum joint error under the shared hypothesis space. Specifically, the target error can be bounded as:

$$\begin{aligned}\epsilon_T(h) &\leq \epsilon_S(h) + \frac{1}{2} d_{\mathcal{H} \Delta \mathcal{H}}(D_S, D_T) + \lambda \\ \lambda &= \min_{h \in \mathcal{H}} (\epsilon_S(h) + \epsilon_T(h))\end{aligned}\tag{S1}$$

where h denotes a specific hypothesis drawn from the hypothesis space $\mathcal{H}$, $\epsilon_S(h)$ denotes the source error, $d_{\mathcal{H} \Delta \mathcal{H}}(D_S, D_T)$ quantifies the distributional discrepancy between the source and target domains, and λ reflects the minimum joint error achievable by the optimal hypothesis under the shared hypothesis space $\mathcal{H}$. When λ is small, a hypothesis exists in $\mathcal{H}$ that fits both domains well, indicating shared structure between the domains and that positive transfer is more likely to occur. Conversely, when λ is large, no hypothesis in $\mathcal{H}$ can simultaneously fit both domains, suggesting a weak theoretical foundation for transfer. This domain adaptation theory suggests that learning performance on the target domain depends not only on the quality of source knowledge, but also on the distributional discrepancy between the two domains and on their intrinsic consistency under a shared representation. Accordingly, positive KT is more likely to occur when the tasks exhibit certain underlying relatedness, high-quality transferred knowledge, and reduced cross-domain discrepancy. Analogously, with respect to EMTO, the source error and target error can be regarded as the fitness values of the source and target tasks. In general, the smaller the target error, the more positive the KT performance.

Inspired by this theoretical perspective, the positive KT of MTLKT can be understood from the following aspects. First, EMTO generally presupposes a certain degree of inter-task similarity, which is analogous to the joint error λ in Eq. (S1). When two tasks show low similarity and differ fundamentally in terms of optimal solution structure or search directionality, the optimal joint error under a shared hypothesis space may still be quite large, causing a high risk of negative transfer. However, even for tasks with low similarity and high joint error λ , MTLKT still tries to achieve positive KT as much as possible. MTLKT establishes a smoother inter-population correspondence by sampling a series of intermediate subspaces along the geodesic flow between the source and target populations in the manifold space, which is intended to mitigate the source-target distributional discrepancy, thereby contributing to a reduction in $d_{\mathcal{H}(\Delta)\mathcal{H}}(D_S, D_T)$. Besides, MTLKT only transfers the potential geometric distribution of the superior individuals from the source population, which is expected to partially mitigate the risk associated with the source error term $\epsilon_S(h)$. In addition, MTLKT further facilitates positive KT from a practical perspective by using the elite selection strategy, in which only the transferred offspring that show superior performance in the target population are retained, whereas inferior ones are excluded from subsequent evolution.

Overall, the design of MTLKT exhibits a certain correspondence with the three key conditions for reducing target error or promoting positive KT, meanwhile, the elite selection strategy also avoids negative KT to some degree in practical perspective. Therefore, MTLKT tries to achieve positive KT as much as possible.

II. EXPERIMENTAL STUDIES

A. EXPERIMENTAL RESULTS ON MULTITASK BENCHMARKS

TABLE S.I PROPERTIES OF THE CEC2017 MULTITASK BENCHMARK

Problem	Task	Search Space	Global Optima	Degree of Intersection	Inter-task Similarity
CI + HS	T_1 : Griewank	$[-100, 100]^{50}$	$\{0\}^{50}$	Complete Intersection	1.0000
	T_2 : Rastrigin	$[-50, 50]^{50}$	$\{0\}^{50}$		
CI + MS	T_1 : Ackley	$[-50, 50]^{50}$	$\{0\}^{50}$	Complete Intersection	0.2261
	T_2 : Rastrigin	$[-50, 50]^{50}$	$\{0\}^{50}$		
CI + LS	T_1 : Ackley	$[-50, 50]^{50}$	$\{42.0969\}^{50}$	Complete Intersection	0.0002
	T_2 : Schwefel	$[-500, 500]^{50}$	$\{420.9687\}^{50}$		
PI + HS	T_1 : Rastrigin	$[-50, 50]^{50}$	$\{0\}^{50}$	Partial Intersection	0.8670
	T_2 : Sphere	$[-100, 100]^{50}$	$\{0\}^{25} \times \{20\}^{25}$		
PI + MS	T_1 : Ackley	$[-50, 50]^{50}$	$\{0\}^{25} \times \{1\}^{25}$	Partial Intersection	0.2152
	T_2 : Rosenbrock	$[-50, 50]^{50}$	$\{1\}^{50}$		
PI + LS	T_1 : Ackley	$[-50, 50]^{50}$	$\{0\}^{50}$	Partial Intersection	0.0725
	T_2 : Weierstrass	$[-0.5, 0.5]^{25}$	$\{0\}^{25}$		
NI + HS	T_1 : Rosenbrock	$[-50, 50]^{50}$	$\{1\}^{50}$	No Intersection	0.9434
	T_2 : Rastrigin	$[-50, 50]^{50}$	$\{0\}^{50}$		
NI + MS	T_1 : Griewank	$[-100, 100]^{50}$	$\{10\}^{50}$	No Intersection	0.3669
	T_2 : Weierstrass	$[-0.5, 0.5]^{50}$	$\{0\}^{50}$		
NI + LS	T_1 : Rastrigin	$[-50, 50]^{50}$	$\{0\}^{50}$	No Intersection	0.0016
	T_2 : Schwefel	$[-500, 500]^{50}$	$\{420.9687\}^{50}$		

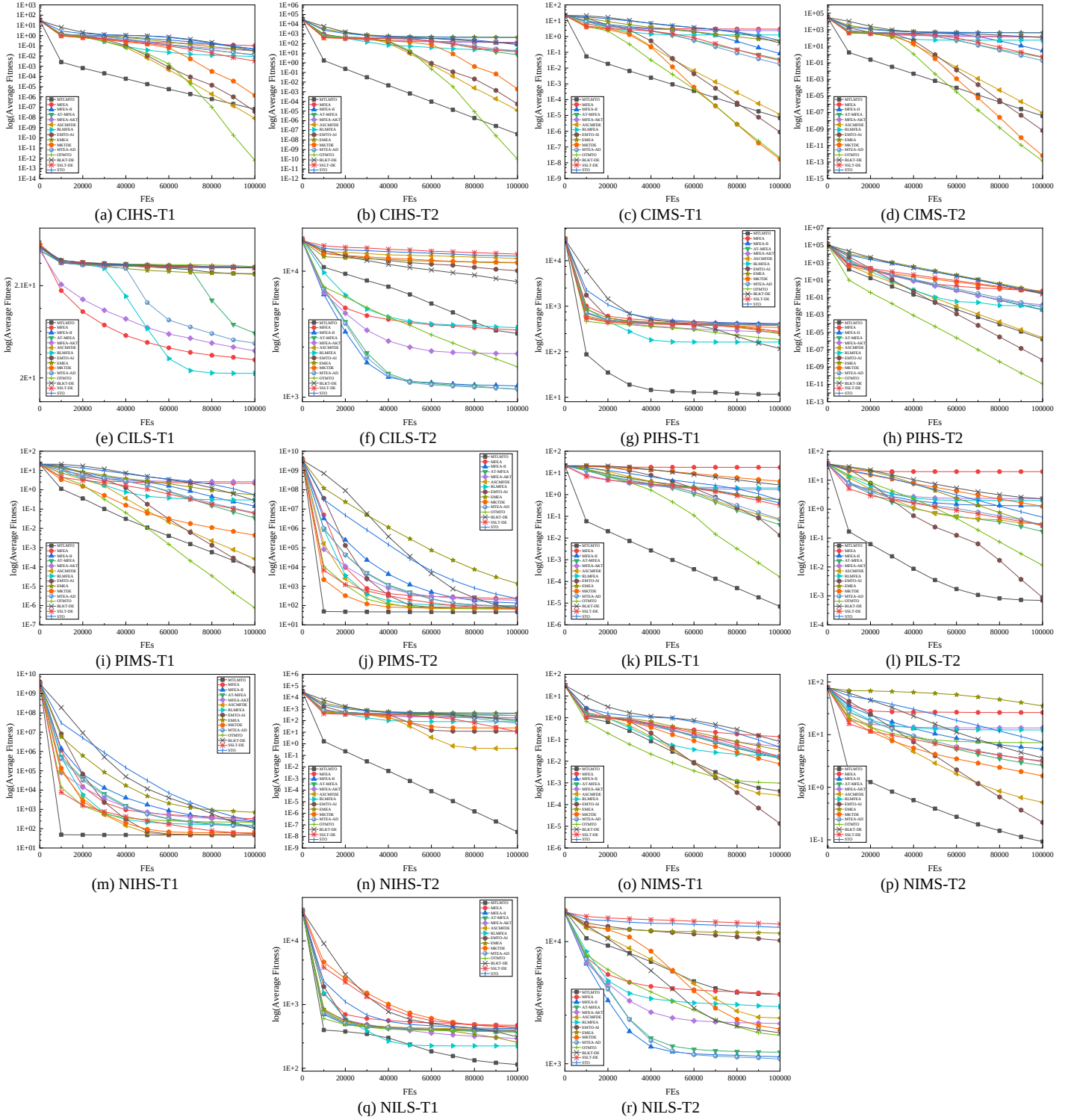


Fig. S1. Convergence curves of different algorithms on the CEC2017 multitask benchmark.

B. EXTENSION EXPERIMENTS ON MANY-TASK BENCHMARKS

TABLE S.II FUNCTION COMBINATION OF EACH PROBLEM ON THE CEC2019 AND WCCI2020 MANY-TASK BENCHMARKS

Function	CEC2019						WCCI2020									
	P1	P2	P3	P4	P5	P6	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10
Sphere							√			√						
Rosenbrock	√							√		√		√		√	√	
Ackley		√								√			√	√	√	√
Rastrigin			√						√		√		√	√	√	√
Griewank				√							√	√		√	√	√
Weierstrass					√						√		√	√	√	√
Schwefel						√						√			√	√

C. REAL-WORLD MULTITASK APPLICATION STUDY

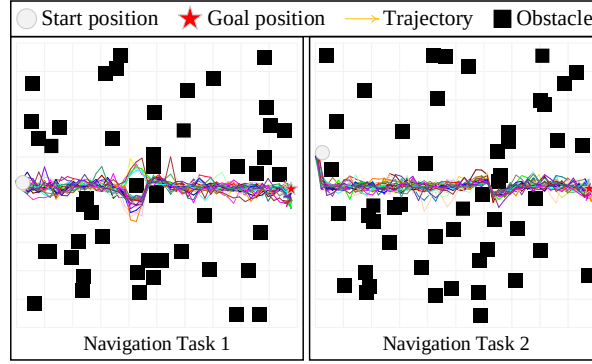


Fig. S3. Navigation trajectories from all 30 independent runs of MTLMT0 for the two tasks in test case C4.

D. PARAMETER SENSITIVITY ANALYSIS

TABLE S.III RESULTS OF MTLMT0 VARIANTS WITH DIFFERENT SETTINGS OF d AND n ON THE CEC2022 MULTITASK

BENCHMARK								
Problem	MTLMT0	$(d, n) = (1, 3)$	$(d, n) = (1, 7)$	$(d, n) = (1, 9)$	$(d, n) = (2, 3)$	$(d, n) = (2, 5)$	$(d, n) = (2, 7)$	$(d, n) = (2, 9)$
P1-T1	6.05E+2	6.05E+2(≈)	6.05E+2(≈)	6.05E+2(≈)	6.04E+2(−)	6.04E+2(−)	6.04E+2(−)	6.04E+2(−)
P1-T2	6.04E+2	6.04E+2(≈)	6.05E+2(≈)	6.05E+2(≈)	6.03E+2(−)	6.03E+2(−)	6.04E+2(≈)	6.04E+2(≈)
P2-T1	7.00E+2	7.00E+2(≈)	7.00E+2(≈)	7.00E+2(≈)	7.00E+2(≈)	7.00E+2(≈)	7.00E+2(≈)	7.00E+2(≈)
P2-T2	7.00E+2	7.00E+2(≈)	7.00E+2(≈)	7.00E+2(≈)	7.00E+2(≈)	7.00E+2(≈)	7.00E+2(≈)	7.00E+2(≈)
P3-T1	2.76E+4	2.57E+4(≈)	2.60E+4(≈)	3.37E+4(≈)	3.90E+4(+)	4.49E+4(+)	4.24E+4(+)	3.24E+4(≈)
P3-T2	4.91E+4	4.27E+4(≈)	4.95E+4(≈)	5.19E+4(≈)	4.78E+4(≈)	6.01E+4(+)	5.82E+4(≈)	5.43E+4(≈)
P4-T1	1.30E+3	1.30E+3(≈)	1.30E+3(≈)	1.30E+3(≈)	1.30E+3(≈)	1.30E+3(≈)	1.30E+3(≈)	1.30E+3(≈)
P4-T2	1.30E+3	1.30E+3(≈)	1.30E+3(≈)	1.30E+3(≈)	1.30E+3(≈)	1.30E+3(≈)	1.30E+3(≈)	1.30E+3(≈)
P5-T1	1.53E+3	1.53E+3(≈)	1.53E+3(≈)	1.53E+3(≈)	1.53E+3(≈)	1.53E+3(≈)	1.53E+3(≈)	1.53E+3(≈)
P5-T2	1.53E+3	1.53E+3(≈)	1.53E+3(≈)	1.53E+3(≈)	1.53E+3(≈)	1.53E+3(≈)	1.53E+3(≈)	1.53E+3(≈)
P6-T1	4.01E+5	4.88E+5(≈)	4.16E+5(≈)	4.55E+5(≈)	5.45E+5(+)	5.13E+5(≈)	5.82E+5(+)	4.61E+5(≈)
P6-T2	2.87E+5	4.38E+5(+)	4.34E+5(+)	3.19E+5(≈)	3.70E+5(+)	4.06E+5(+)	3.79E+5(≈)	4.42E+5(+)
P7-T1	2.60E+3	2.53E+3(≈)	2.65E+3(≈)	2.53E+3(≈)	2.48E+3(≈)	2.53E+3(≈)	2.55E+3(≈)	2.58E+3(≈)
P7-T2	2.56E+3	2.68E+3(≈)	2.64E+3(≈)	2.64E+3(≈)	2.65E+3(≈)	2.59E+3(≈)	2.59E+3(≈)	2.65E+3(≈)
P8-T1	5.21E+2	5.21E+2(≈)	5.21E+2(≈)	5.21E+2(≈)	5.21E+2(≈)	5.21E+2(≈)	5.21E+2(≈)	5.21E+2(≈)

P8-T2	5.21E+2	5.21E+2($\approx$)	5.21E+2($\approx$)	5.21E+2($\approx$)	5.21E+2(+)	5.21E+2(+)	5.21E+2($\approx$)	5.21E+2(+)
P9-T1	6.88E+3	6.94E+3($\approx$)	7.07E+3($\approx$)	6.98E+3($\approx$)	7.13E+3($\approx$)	6.94E+3($\approx$)	7.06E+3($\approx$)	6.99E+3($\approx$)
P9-T2	1.62E+3	1.62E+3($\approx$)	1.62E+3($\approx$)	1.62E+3($\approx$)	1.62E+3(+)	1.62E+3($\approx$)	1.62E+3($\approx$)	1.62E+3(+)
P10-T1	9.72E+3	9.19E+3($\approx$)	9.98E+3($\approx$)	9.65E+3($\approx$)	1.03E+4($\approx$)	9.32E+3($\approx$)	1.09E+4($\approx$)	1.09E+4($\approx$)
P10-T2	6.81E+5	6.52E+5($\approx$)	7.33E+5($\approx$)	7.86E+5($\approx$)	6.84E+5($\approx$)	8.23E+5($\approx$)	7.86E+5($\approx$)	7.29E+5($\approx$)
+ / $\approx$ / -	$\sim$	1 / 19 / 0	1 / 19 / 0	0 / 20 / 0	5 / 13 / 2	4 / 14 / 2	2 / 17 / 1	3 / 16 / 1
Problem	$(d, n) = (3, 3)$	$(d, n) = (3, 5)$	$(d, n) = (3, 7)$	$(d, n) = (3, 9)$	$(d, n) = (4, 3)$	$(d, n) = (4, 5)$	$(d, n) = (4, 7)$	$(d, n) = (4, 9)$
P1-T1	6.03E+2(-)	6.04E+2(-)	6.04E+2(-)	6.04E+2(-)	6.03E+2(-)	6.03E+2(-)	6.03E+2(-)	6.03E+2(-)
P1-T2	6.04E+2(-)	6.04E+2(-)	6.04E+2($\approx$)	6.04E+2($\approx$)	6.03E+2(-)	6.03E+2(-)	6.03E+2(-)	6.03E+2(-)
P2-T1	7.00E+2($\approx$)	7.00E+2($\approx$)	7.00E+2($\approx$)	7.00E+2($\approx$)	7.00E+2($\approx$)	7.00E+2($\approx$)	7.00E+2($\approx$)	7.00E+2($\approx$)
P2-T2	7.00E+2($\approx$)	7.00E+2($\approx$)	7.00E+2($\approx$)	7.00E+2($\approx$)	7.00E+2($\approx$)	7.00E+2($\approx$)	7.00E+2($\approx$)	7.00E+2($\approx$)
P3-T1	5.33E+4(+)	5.62E+4(+)	5.65E+4(+)	5.76E+4(+)	8.30E+4(+)	1.15E+5(+)	1.02E+5(+)	7.88E+4(+)
P3-T2	7.85E+4(+)	7.96E+4(+)	8.75E+4(+)	7.98E+4(+)	1.11E+5(+)	1.17E+5(+)	1.23E+5(+)	1.25E+5(+)
P4-T1	1.30E+3(+)	1.30E+3($\approx$)	1.30E+3($\approx$)	1.30E+3($\approx$)	1.30E+3($\approx$)	1.30E+3($\approx$)	1.30E+3($\approx$)	1.30E+3($\approx$)
P4-T2	1.30E+3($\approx$)	1.30E+3($\approx$)	1.30E+3($\approx$)	1.30E+3($\approx$)	1.30E+3($\approx$)	1.30E+3($\approx$)	1.30E+3($\approx$)	1.30E+3($\approx$)
P5-T1	1.53E+3($\approx$)	1.53E+3($\approx$)	1.53E+3($\approx$)	1.53E+3($\approx$)	1.53E+3($\approx$)	1.53E+3($\approx$)	1.53E+3($\approx$)	1.53E+3($\approx$)
P5-T2	1.53E+3($\approx$)	1.53E+3($\approx$)	1.53E+3($\approx$)	1.53E+3($\approx$)	1.53E+3($\approx$)	1.53E+3($\approx$)	1.53E+3($\approx$)	1.53E+3($\approx$)
P6-T1	5.59E+5(+)	5.39E+5(+)	5.26E+5(+)	5.16E+5($\approx$)	5.98E+5(+)	7.31E+5(+)	5.47E+5(+)	6.66E+5(+)
P6-T2	3.89E+5($\approx$)	4.24E+5($\approx$)	3.53E+5($\approx$)	5.04E+5(+)	4.35E+5(+)	4.76E+5(+)	5.18E+5(+)	6.09E+5(+)
P7-T1	2.70E+3($\approx$)	2.59E+3($\approx$)	2.54E+3($\approx$)	2.48E+3($\approx$)	2.60E+3($\approx$)	2.50E+3($\approx$)	2.55E+3($\approx$)	2.51E+3($\approx$)
P7-T2	2.61E+3($\approx$)	2.71E+3(+)	2.60E+3($\approx$)	2.58E+3($\approx$)	2.68E+3(+)	2.72E+3($\approx$)	2.63E+3($\approx$)	2.71E+3(+)
P8-T1	5.21E+2($\approx$)	5.21E+2($\approx$)	5.21E+2($\approx$)	5.21E+2(+)	5.21E+2($\approx$)	5.21E+2($\approx$)	5.21E+2($\approx$)	5.21E+2($\approx$)
P8-T2	5.21E+2($\approx$)	5.21E+2(+)	5.21E+2($\approx$)	5.21E+2($\approx$)	5.21E+2($\approx$)	5.21E+2($\approx$)	5.21E+2(+)	5.21E+2($\approx$)
P9-T1	7.53E+3($\approx$)	7.60E+3(+)	7.09E+3($\approx$)	7.25E+3($\approx$)	8.12E+3(+)	8.07E+3(+)	8.17E+3(+)	8.11E+3(+)
P9-T2	1.62E+3(+)	1.62E+3(+)	1.62E+3(+)	1.62E+3(+)	1.62E+3(+)	1.62E+3(+)	1.62E+3(+)	1.62E+3(+)
P10-T1	9.69E+3($\approx$)	9.60E+3($\approx$)	1.14E+4($\approx$)	1.08E+4($\approx$)	1.10E+4($\approx$)	1.15E+4($\approx$)	1.23E+4(+)	1.01E+4($\approx$)
P10-T2	9.37E+5($\approx$)	8.70E+5(+)	1.05E+6(+)	8.43E+5($\approx$)	7.74E+5($\approx$)	7.78E+5($\approx$)	8.77E+5(+)	7.12E+5($\approx$)
+ / $\approx$ / -	5 / 13 / 2	8 / 10 / 2	5 / 14 / 1	5 / 14 / 1	7 / 11 / 2	6 / 12 / 2	9 / 9 / 2	7 / 11 / 2

To further illustrate the rationality of $n = 5$, we have analyzed the impact of n from a theoretical aspect according to [S3], which integrates over all subspaces along the entire geodesic continuously, corresponding to the limiting case $n \rightarrow \infty$ in our discrete sampling scheme, and thus serves as a theoretical upper bound reference on the geometric information that our discrete sampling scheme can extract. Specifically, for two samples $\mathbf{x}_i$ and $\mathbf{x}_j$, their projections into $\Phi(t)$ for a continuous t from 0 to 1 are concatenated into infinite-dimensional feature vectors $\mathbf{z}_i^\infty$ and $\mathbf{z}_j^\infty$, respectively. The inner product between them defines the geodesic flow kernel:

$$\langle \mathbf{z}_i^\infty, \mathbf{z}_j^\infty \rangle = \int_0^1 (\Phi(t)^\top \mathbf{x}_i)^\top (\Phi(t)^\top \mathbf{x}_j) dt = \mathbf{x}_i^\top \mathbf{G} \mathbf{x}_j \quad (\text{S3})$$

where $\mathbf{G} \in \mathbb{R}^{D \times D}$ is a symmetric positive semidefinite matrix with the following closed-form solution [S3]:

$$\mathbf{G} = [\mathbf{P}_s \mathbf{U}_1 \mathbf{R}_s \mathbf{U}_2] \begin{bmatrix} \Lambda_1 & \Lambda_2 \\ \Lambda_2 & \Lambda_3 \end{bmatrix} \begin{bmatrix} \mathbf{U}_1^\top \mathbf{P}_s^\top \\ \mathbf{U}_2^\top \mathbf{R}_s^\top \end{bmatrix} \quad (\text{S4})$$

where $\Lambda_1, \Lambda_2, \Lambda_3$ are diagonal matrices whose i^{th} diagonal elements are:

$$\lambda_{1i} = 1 + \frac{\sin(2\theta_i)}{2\theta_i}, \quad \lambda_{2i} = \frac{\cos(2\theta_i) - 1}{2\theta_i}, \quad \lambda_{3i} = 1 - \frac{\sin(2\theta_i)}{2\theta_i} \quad (\text{S5})$$

Since $\mathbf{G}$ is an integral of outer products of the form $\Phi(t)\Phi(t)^T$, it is symmetric positive semidefinite by construction, and thus admits a symmetric matrix square root $\mathbf{G}^{1/2}$. The inner product can therefore be rewritten as:

$$\mathbf{x}_i^T \mathbf{G} \mathbf{x}_j = \mathbf{x}_i^T \mathbf{G}^{1/2} \mathbf{G}^{1/2} \mathbf{x}_j = (\mathbf{G}^{1/2} \mathbf{x}_i)^T (\mathbf{G}^{1/2} \mathbf{x}_j) = \xi_i^T \xi_j \quad (\text{S6})$$

where ξ denotes the explicit feature representation of sample $\mathbf{x}$, given by:

$$\xi = \mathbf{G}^{1/2} \mathbf{x} \quad (\text{S7})$$

Eq. (S7) represents the explicit feature mapping under the continuous subspace, which directly corresponds to the discrete subspace sampling mapping in Eq. (S2). Herein, we construct a variant MTLMT0-CS by replacing the discrete feature construction in Eq. (S2) with the explicit feature mapping in Eq. (S7), and compare MTLMT0-CS against the original MTLMT0 with $n = 5$ on the CEC2022 multitask benchmark. The comparison results are reported in Table S.IV. As shown in Table S.IV, we find that MTLMT0 and MTLMT0-CS achieve similar performance across all 20 tasks, indicating that the discrete sampling scheme with $n = 5$ is able to extract sufficient geometric information close to the theoretical upper bound reference. Therefore, the choice of $n = 5$ is a reasonable and effective choice, supported by both the theoretical analysis above and the empirical evidence presented.

TABLE S.IV RESULTS OF MTLMT0 AND MTLMT0-CS ON THE CEC2022 MULTITASK BENCHMARK

Problem	MTLMT0	MTLMT0-CS
P1-T1	6.05E+2	6.04E+2 ($\approx$)
P1-T2	6.04E+2	6.04E+2 ($\approx$)
P2-T1	7.00E+2	7.00E+2 ($\approx$)
P2-T2	7.00E+2	7.00E+2($\approx$)
P3-T1	2.76E+4	2.70E+4 ($\approx$)
P3-T2	4.91E+4	3.87E+4 ($\approx$)
P4-T1	1.30E+3	1.30E+3 ($\approx$)
P4-T2	1.30E+3	1.30E+3 ($\approx$)
P5-T1	1.53E+3	1.53E+3 ($\approx$)
P5-T2	1.53E+3	1.53E+3($\approx$)
P6-T1	4.01E+5	4.25E+5($\approx$)
P6-T2	2.87E+5	3.56E+5($\approx$)
P7-T1	2.60E+3	2.59E+3 ($\approx$)
P7-T2	2.56E+3	2.59E+3($\approx$)
P8-T1	5.21E+2	5.21E+2 ($\approx$)
P8-T2	5.21E+2	5.21E+2($\approx$)
P9-T1	6.88E+3	6.98E+3($\approx$)
P9-T2	1.62E+3	1.62E+3($\approx$)
P10-T1	9.72E+3	9.73E+3($\approx$)
P10-T2	6.81E+5	7.92E+5($\approx$)
Number of + / $\approx$ / -		0 / 20 / 0

TABLE S.V RESULTS OF MTLMT0 VARIANTS WITH DIFFERENT BCR ON THE CEC2022 MULTITASK BENCHMARK

Problem	MTLMT0	$BCR = 0.8$	$BCR = 0.7$	$BCR = 0.6$	$BCR = 0.5$	$BCR = 0.4$	$BCR = 0.3$	$BCR = 0.2$	$BCR = 0.1$
P1-T1	6.05E+2	6.03E+2(−)	6.03E+2(−)	6.03E+2(−)	6.03E+2(−)	6.03E+2(−)	6.03E+2(−)	6.03E+2(−)	6.04E+2(≈)
P1-T2	6.04E+2	6.03E+2(−)	6.03E+2(−)	6.03E+2(−)	6.03E+2(−)	6.03E+2(−)	6.03E+2(−)	6.03E+2(−)	6.04E+2(≈)
P2-T1	7.00E+2	7.00E+2(+)	7.00E+2(+)	7.00E+2(+)	7.00E+2(+)	7.00E+2(+)	7.00E+2(+)	7.00E+2(+)	7.00E+2(≈)
P2-T2	7.00E+2	7.00E+2(+)	7.00E+2(+)	7.00E+2(+)	7.00E+2(+)	7.00E+2(+)	7.00E+2(+)	7.00E+2(+)	7.00E+2(+)
P3-T1	2.76E+4	9.39E+4(+)	1.50E+5(+)	1.89E+5(+)	1.91E+5(+)	1.92E+5(+)	1.63E+5(+)	1.12E+5(+)	4.46E+4(+)
P3-T2	4.91E+4	1.70E+5(+)	2.42E+5(+)	3.71E+5(+)	3.84E+5(+)	3.68E+5(+)	2.94E+5(+)	2.05E+5(+)	7.47E+4(+)
P4-T1	1.30E+3	1.30E+3(+)	1.30E+3(+)	1.30E+3(+)	1.30E+3(+)	1.30E+3(+)	1.30E+3(+)	1.30E+3(+)	1.30E+3(+)
P4-T2	1.30E+3	1.30E+3(+)	1.30E+3(+)	1.30E+3(+)	1.30E+3(+)	1.30E+3(+)	1.30E+3(+)	1.30E+3(+)	1.30E+3(+)
P5-T1	1.53E+3	1.53E+3(+)	1.53E+3(+)	1.53E+3(+)	1.53E+3(+)	1.53E+3(+)	1.53E+3(+)	1.53E+3(+)	1.53E+3(+)
P5-T2	1.53E+3	1.54E+3(+)	1.54E+3(+)	1.54E+3(+)	1.54E+3(+)	1.54E+3(+)	1.54E+3(+)	1.53E+3(+)	1.53E+3(+)
P6-T1	4.01E+5	1.53E+6(+)	2.82E+6(+)	3.01E+6(+)	3.75E+6(+)	3.28E+6(+)	3.00E+6(+)	1.78E+6(+)	6.92E+5(+)
P6-T2	2.87E+5	1.93E+6(+)	3.23E+6(+)	4.75E+6(+)	5.22E+6(+)	5.33E+6(+)	4.15E+6(+)	2.11E+6(+)	6.79E+5(+)
P7-T1	2.60E+3	2.84E+3(+)	3.24E+3(+)	3.46E+3(+)	3.49E+3(+)	3.48E+3(+)	3.40E+3(+)	3.01E+3(+)	2.54E+3(≈)
P7-T2	2.56E+3	2.79E+3(+)	3.24E+3(+)	3.29E+3(+)	3.52E+3(+)	3.24E+3(+)	3.27E+3(+)	2.93E+3(+)	2.57E+3(≈)
P8-T1	5.21E+2	5.21E+2(≈)	5.21E+2(≈)	5.21E+2(≈)	5.21E+2(≈)	5.21E+2(≈)	5.21E+2(≈)	5.21E+2(≈)	5.21E+2(≈)
P8-T2	5.21E+2	5.21E+2(≈)	5.21E+2(≈)	5.21E+2(≈)	5.21E+2(+)	5.21E+2(≈)	5.21E+2(≈)	5.21E+2(≈)	5.21E+2(≈)
P9-T1	6.88E+3	6.71E+3(≈)	7.58E+3(≈)	1.01E+4(+)	1.23E+4(+)	1.06E+4(+)	7.96E+3(≈)	6.52E+3(≈)	6.69E+3(≈)
P9-T2	1.62E+3	1.62E+3(+)	1.62E+3(+)	1.62E+3(+)	1.62E+3(+)	1.62E+3(+)	1.62E+3(+)	1.62E+3(+)	1.62E+3(≈)
P10-T1	9.72E+3	1.46E+4(+)	1.61E+4(+)	1.72E+4(+)	1.79E+4(+)	1.77E+4(+)	1.67E+4(+)	1.42E+4(+)	1.17E+4(+)
P10-T2	6.81E+5	2.19E+6(+)	3.53E+6(+)	4.48E+6(+)	5.42E+6(+)	4.94E+6(+)	4.39E+6(+)	2.88E+6(+)	1.17E+6(+)
Number of + / ≈ / −		15 / 3 / 2	15 / 3 / 2	16 / 2 / 2	17 / 1 / 2	16 / 2 / 2	15 / 3 / 2	15 / 3 / 2	11 / 9 / 0

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